Physics 5A Midterm 1 Review Session

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Use $g = 9.8\,\mathrm{m/s^2}$ and neglect air resistance in problems 1 and 2. Make sure you show your work and include units in your final answers.

Problem 1: Jinyan throws an eraser directly upward with an initial speed $5.00\,\mathrm{m/s}$. The eraser travels up and then comes back down. Jinyan catches it on its way down, and its speed just before he catches it is $6.00\,\mathrm{m/s}$. Take the positive y-direction to be upward.

Part (a): Find the vertical displacement of the eraser from the moment it leaves Jinyan's hand to the moment just before he catches it. Does he catch it above or below its release point?

Part (b): Find the maximal height the eraser can reach, measured with respect to its release point.

Part (c): Find the total time the eraser spends in the air before Jinyan catches it.

Part (d): Find the total distance that the eraser has traveled in the air.

Problem 2: During the discussion this week, Jinyan threw an eraser into the air at an angle $45°$ above the horizontal, toward the right. The eraser hit the ground 1.00 second after it left his hand. Its vertical displacement from the release point to the ground was $-1.40\,\mathrm{m}$. Take the positive x-direction to be to the right, and the positive y-direction to be upward.

Part (a): Find the y-component of the eraser's initial velocity.

Part (b): Find the x-component of the eraser's initial velocity.

Part (c): Find the x-component of the eraser's velocity when it reaches the maximal height of its trajectory.

Problem 3: A rocket in space has two thrusters. The first thruster provides an acceleration $\vec{a}_1$ with magnitude $6.00\,\mathrm{m/s^2}$ at an angle 30° above the positive x-direction. The second thruster provides an acceleration $\vec{a}_2$ with magnitude $2\sqrt{3}\,\mathrm{m/s^2}$ at an angle 60° below the positive x-direction. The total acceleration of the rocket is the vector sum of these two accelerations. Neglect gravity and any other forces in this problem.

Part (a): Find the x- and y-components of $\vec{a}_1$ and $\vec{a}_2$. Then find the total acceleration $\vec{a} = \vec{a}_1 + \vec{a}_2$, written in terms of $\hat{x}$ and $\hat{y}$.

Part (b): Find the magnitude and direction of the total acceleration $\vec{a}$. Write its direction as an angle measured from the positive x-direction.

Part (c): Now the magnitude of $\vec{a}_2$ is reduced by half, while its direction and the entire vector $\vec{a}_1$ stay the same. Find the components of the new acceleration from the second thruster, then compute the new total acceleration vector $\vec{a}\,'$.

Part (d): Find the magnitude and direction of $\vec{a}\,'$. Does the total acceleration still point in the positive x-direction? Explain your answer.